



SVCE KADAPA
EDUCATION FOR A BETTER SOCIETY

MACHINE LEARNING IN MONEY LAUNDERING DETECTION USING CRYPTOCURRENCY DATA

PRESENTED BY

S.Shoaib(22KH1A3340)
C.Harinath(22KH1A3310)
D.Vyshnavi(22KH1A3312)

GUIDED BY

D.Masthan Pasha ,M.Tech.,(ph.D)
Assistant Professor Dept of CSE(AI&ML)

CONTENTS

- Abstract
- Introduction
- Objectives
- Existing System
- Disadvantages
- Proposed System
- Advantages
- System Requirements
- System Architecture
- Modules
- UML Diagrams
- Test Cases
- Output Screenshots
- Conclusion
- Future Enhancement

ABSTRACT

Money laundering is a major financial crime that hides illegal funds through complex transaction networks. This project uses machine learning techniques to analyze cryptocurrency transaction data and detect suspicious patterns. The system identifies unusual activities that may indicate money laundering using the XGBoost algorithm, which effectively handles complex data. It enables faster and more accurate detection of financial crimes while reducing false positives. The model achieves around 98% accuracy, improving the overall efficiency and reliability of the system.

INTRODUCTION

The growth of cryptocurrency has improved digital transactions but also increased financial crimes due to user anonymity. Traditional AML systems are inflexible and struggle to handle complex and evolving transaction patterns. Machine learning enables accurate detection by analyzing large-scale data and learning transaction behavior. It adapts to new laundering techniques, reduces false positives, and improves overall security and efficiency in cryptocurrency systems.

OBJECTIVES

- Detect suspicious wallets and transactions by analyzing unusual patterns, high-risk behavior, and abnormal fund movements.
- Reduce AML false positives by improving accuracy and filtering out genuine transactions from risky ones.
- Support regulators by providing transparent monitoring, reports, and compliance-ready transaction tracking.
- Identify layered money laundering activities by tracing complex transaction chains across multiple wallets.

EXISTING SYSTEM

Existing money laundering detection systems mainly use traditional models like Support Vector Machines (SVM). These models analyze basic features such as transaction amount, frequency, and wallet activity, achieving around 90% accuracy on simple datasets and offering good computational efficiency. However, they struggle with complex, non-linear, and evolving transaction patterns. Advanced techniques like chaining and mixing are not effectively captured, leading to reduced real-world accuracy and higher false positives, making them less suitable for modern cryptocurrency environments.

DISADVANTAGES

High Computational Cost

High False Positive
Rates

Higher Training Time for
Large Datasets

Limited Scalability for
Large Data

PROPOSED SYSTEM

The proposed system uses XGBoost, an advanced ensemble learning model, to detect money laundering in cryptocurrency transactions. It combines multiple decision trees to improve accuracy and handles complex, non-linear data effectively. The model reduces overfitting, processes large volumes of data efficiently, and adapts to new patterns through continuous learning. This results in higher detection accuracy and fewer false positives compared to traditional methods, making it suitable for modern cryptocurrency environments.

ADVANTAGES

High Accuracy

Reduce False Positives

Scalable for Larger
Data

Handles Missing Data

SYSTEM REQUIREMENTS

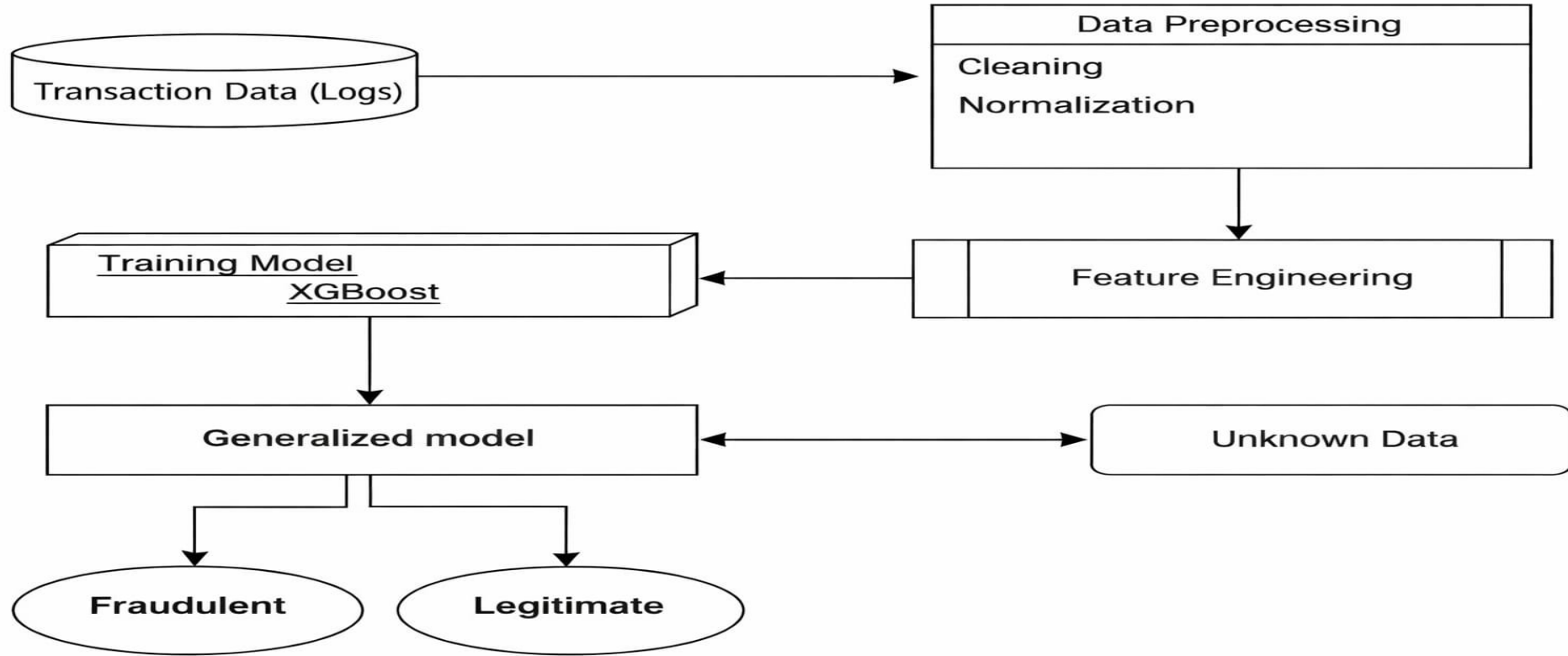
Software Requirements

- Windows / Linux OS
- Python 3.8+
- Flask Framework
- NumPy, Pandas, Scikit-learn,
- XGBoost

Hardware Requirements

- Intel i5 or higher processor
- 8 GB RAM minimum
- 100 GB HDD/SSD
- Internet connectivity

SYSTEM ARCHITECTURE



MODULES

- Data Collection
- Data Preprocessing Module
- Feature Extraction Module
- Model Training Module
- Suspicious Transaction Detection Module

DATA COLLECTION MODULE

The Data Collection Module is the first stage of the system, where cryptocurrency transaction data is collected from various platforms. It includes details such as transaction IDs, wallet addresses, timestamps, transaction amounts, and block numbers. In this project, the dataset is obtained from Kaggle and contains 30,001 rows and 9 columns. The main objective is to gather reliable data for training machine learning models, which is then stored in formats like CSV files or databases for further processing.

DATA PREPROCESSING MODULE

After collecting the data, the Data Preprocessing Module is used to clean and organize it. The cryptocurrency transaction data from the Blockchain may contain missing values, duplicates, or irrelevant information. This module removes duplicates, handles missing values, and converts categorical data into numerical form. It also extracts useful features like transaction frequency and transaction volume. Finally, normalization or scaling is applied to improve model performance. This step is important because machine learning models need clean and structured data for accurate predictions.

FEATURE ENGINEERING

The Feature Extraction Module selects important information from preprocessed cryptocurrency data. Since not all data is useful, only meaningful features are chosen to help the model understand transaction behavior. Common features include transaction amount, frequency, number of wallet interactions, and average transaction value. It can also use network relationships between wallets to detect suspicious patterns. This module converts the data into a structured format that can be used by machine learning models.

MODEL TRAINING MODULE

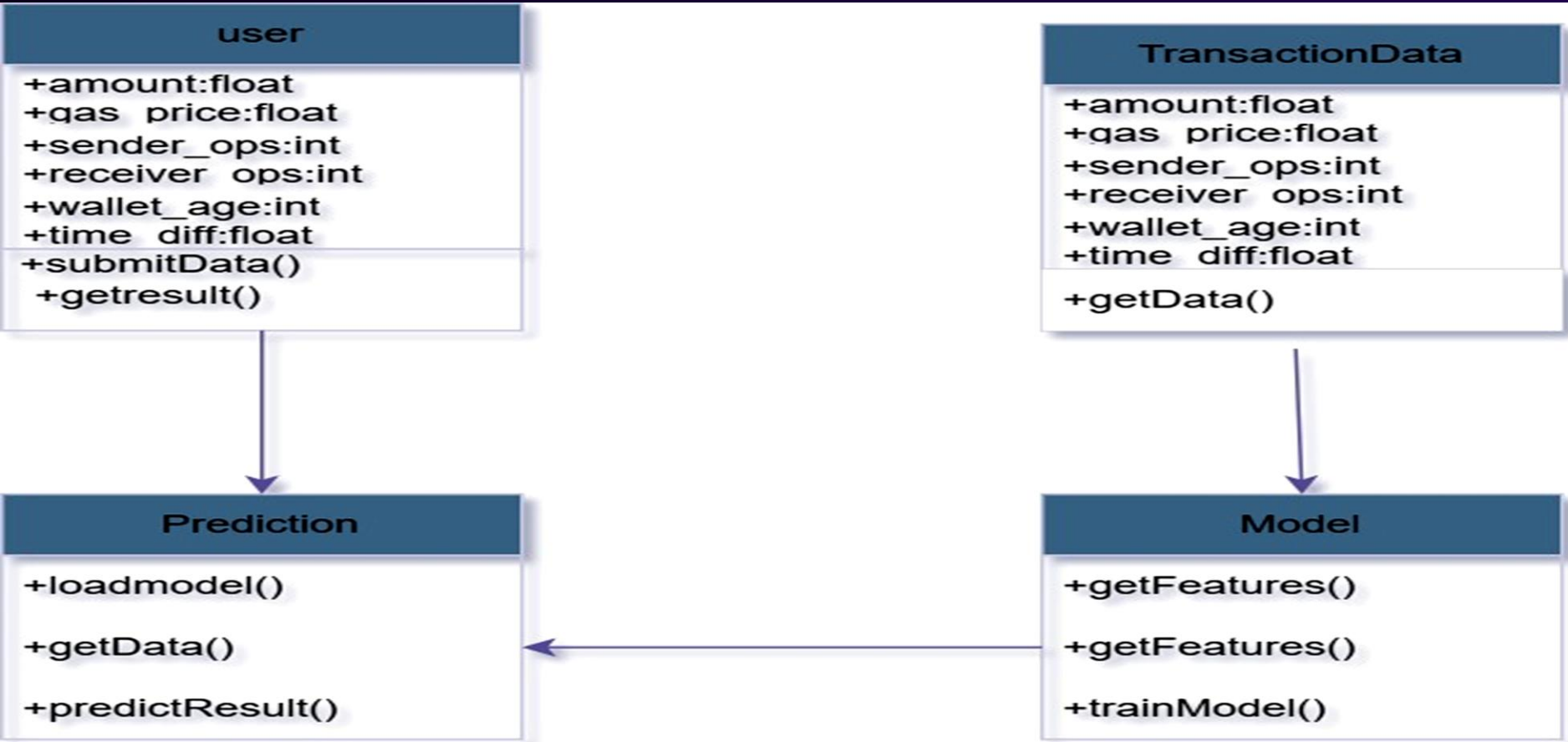
The Model Training Module is where machine learning algorithms are trained using the prepared dataset. Algorithms like Random Forest, XGBoost, Logistic Regression, and Neural Networks are used to detect suspicious transactions. The data is split into training and testing sets. The model learns patterns from the training data and adjusts itself to reduce errors and improve accuracy. The main goal is to build a model that can accurately detect money laundering transactions.

SUSPICIOUS TRANSACTION DETECTION MODULE

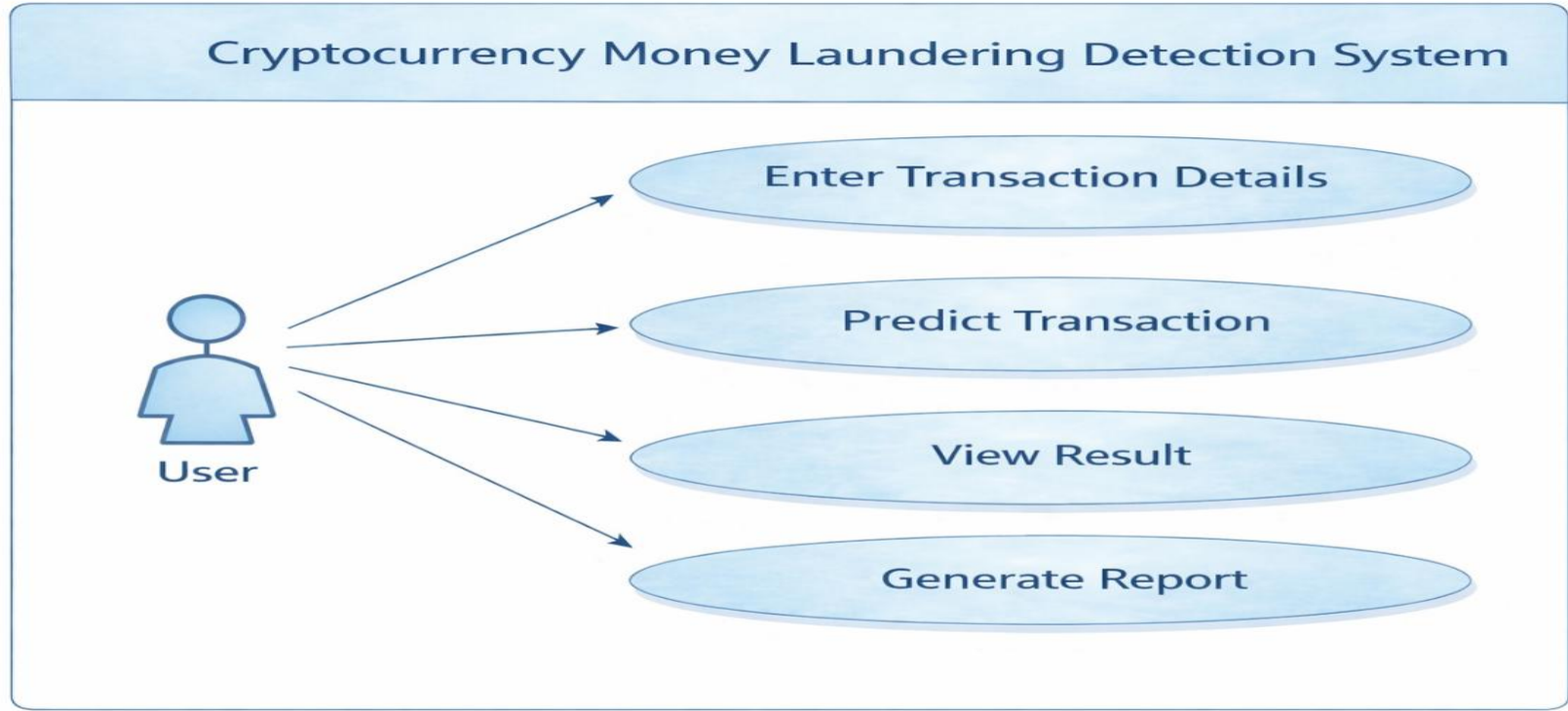
The Suspicious Transaction Detection Module uses the trained model to analyze new cryptocurrency transaction data. It examines each transaction and predicts whether it is normal or suspicious based on learned patterns. If a transaction is flagged as suspicious, it may indicate possible money laundering activity. The system can also generate alerts for further investigation by analysts or authorities. This module helps organizations monitor large volumes of transactions efficiently, reduces manual effort, and improves the speed and accuracy of detecting illegal activities.

UML DIAGRAMS

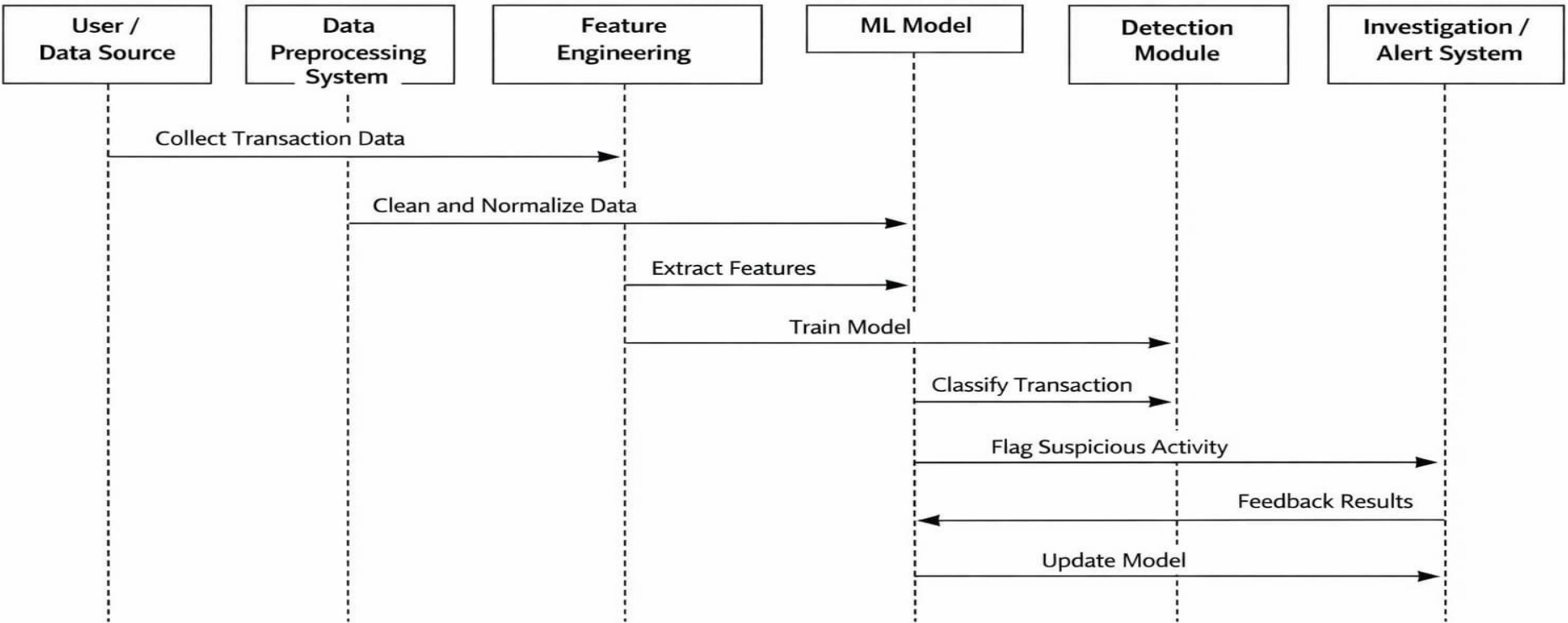
CLASS DIAGRAM



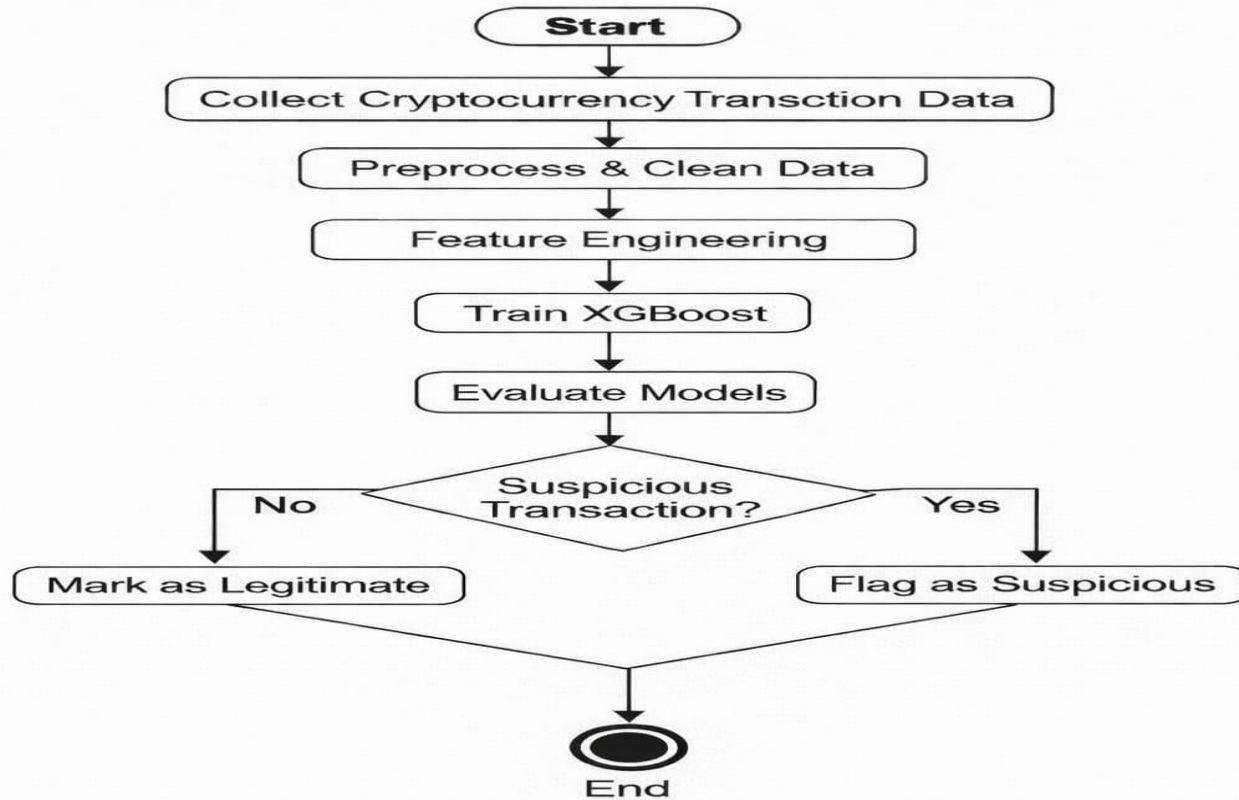
USECASE DIAGRAM



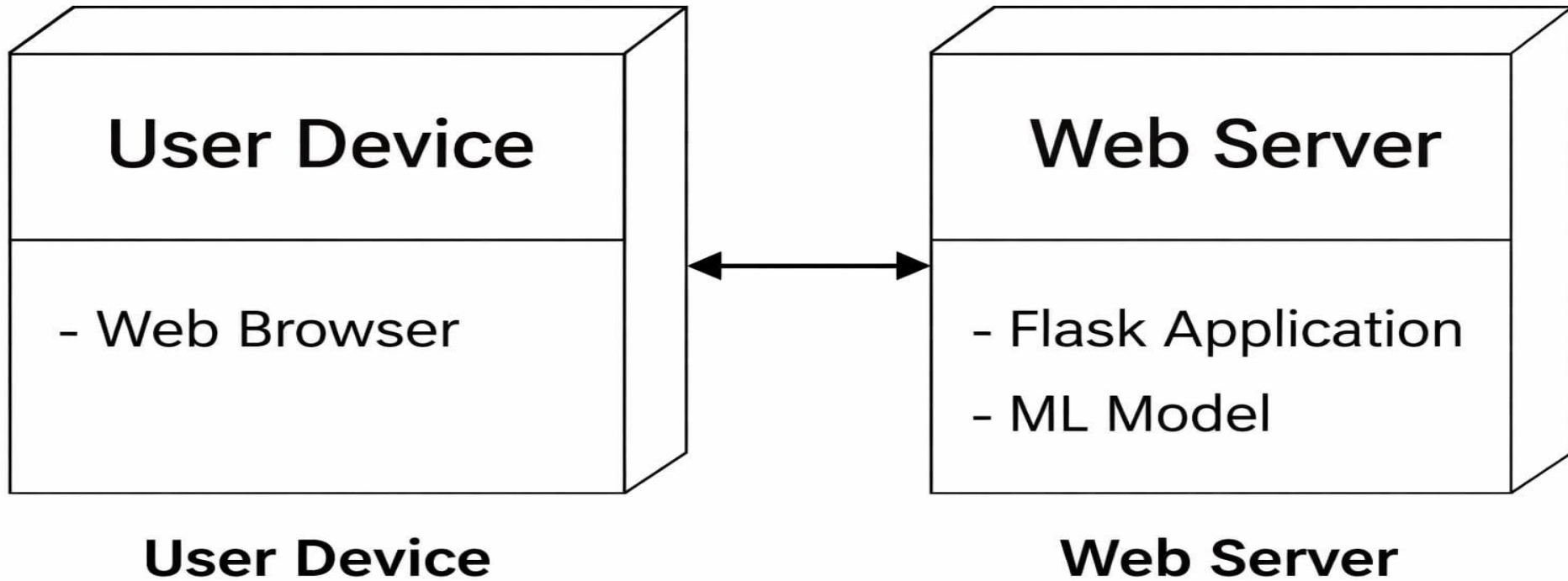
SEQUENCE DIAGRAM



ACTIVITY DIAGRAM



DEPLOYMENT DIAGRAM



TESTING TECHNIQUES

UNIT TESTING

Unit testing focuses on verifying the smallest components of a system individually. In the Cryptocurrency Money Laundering Detection System, modules like data collection, preprocessing, feature extraction, model training, and transaction detection are tested separately. Each function is checked with different inputs to ensure correct outputs. This helps identify errors early and ensures each module works properly, improving overall system reliability.

INTEGRATION TESTING

Integration testing ensures that different modules of the system work together smoothly. In this system, it verifies the connection between data collection, preprocessing, feature extraction, model training, and transaction detection. It checks whether data flows correctly between modules and is properly used at each stage. It also ensures that prediction results are accurately generated and displayed or stored. This helps identify issues related to module interaction, interfaces, and overall system functionality.

TEST CASES

S. No.	TEST CASES	INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
1	Open Money Laundering Detection System	Open the web application in browser	Home page with transaction input form should be displayed	Home page displayed successfully	Pass
2	Transaction Data Submission	Enter transaction details (Transaction ID, Amount, Sender Address, Receiver Address, Time)	System should analyze transaction data	System should analyze transaction data	Pass
3	Missing Input Fields	User leaves required transaction fields empty	System should not process the transaction	Prediction cannot be performed due to missing values	Fail
4	Upload Cryptocurrency Dataset	Upload blockchain transaction dataset	Dataset should be loaded into the system	Dataset loaded successfully	Pass
5	Data Preprocessing	Verify dataset preprocessing	System should clean and prepare the dataset	Dataset preprocessed successfully	Pass
6	Run XGBoost Algorithm	Execute XGBoost model for training	Model should train using the dataset	XGBoost model executed successfully	Pass
7	Detect Suspicious Transaction	Provide transaction data for prediction	System should classify transaction as Normal or Suspicious	Transaction classified successfully	Pass

OUTPUT SCREENSHOTS

CRYPTOGUARD

HomeDashboard

Advanced Crypto Fraud Detection

Harnessing ML to secure every transaction on the blockchain.

Transaction Amount (USD)

454

Gas Price (Gwei)

34

Time Since Last TX (Mins)

454

Sender Total History

35

Receiver Total History

45

Wallet Age (Days)

543

Network Congestion (0-1)

0.6

IP Reputation (0-1)

0.8

Analyze Transaction

Transaction Analysis Result

SAFE TRANSACTION

The transaction parameters align with legitimate blockchain activity.

XAI Suggestion:

Security Check: The transaction shows unusual high gas usage which often indicates a forced or rushed malicious smart contract interaction intent.

Amount

454.0

Time diff

454.0

Receiver ops

45.0

Network congestion

0.6

Gas price

34.0

Sender ops

35.0

Wallet age

543.0

To reputation

0.8

New Analysis

View Dashboard

Model Performance & Dataset Details

30000

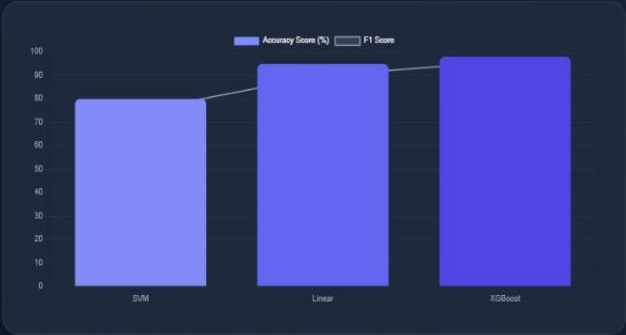
Total Samples

8

Input Features

3

ML Algorithms



Model Comparison Matrix			
Algorithm	Status	Accuracy (Target)	F1 Score
Support Vector Machine (SVM)	Optimized	80%	0.75
Linear Regression (Logistic)	Optimized	95%	0.92
XG Booster (XGB)	Superior	98%	0.97

Transaction Analysis Result

FRAUDULENT DETECTED

Our AI models have identified high-risk patterns in this transaction.

XAI Suggestion:

Warning: Extreme network congestion during this high-value transfer may indicate an attempt to exploit timing delays in decentralized exchange price feeds.

Amount
48495.0

Time diff
1406.0

Receiver ops
432.0

Network congestion
0.42

Gas price
448.0

Sender ops
557.0

Wallet age
3064.0

Ip reputation
0.45

New Analysis

View Dashboard

CONCLUSION

The rapid growth of cryptocurrencies has improved digital transactions by making them fast and decentralized. However, the pseudo-anonymous nature of the CryptoCurrency transaction data also creates opportunities for illegal activities like money laundering.

This project uses machine learning techniques such as Support Vector Machine, Linear Regression, and XGBoost to detect suspicious transactions. The system analyzes features like transaction amount, wallet addresses, and transaction frequency.

In conclusion, the project shows that machine learning can effectively detect money laundering and improve monitoring of cryptocurrency transactions.

FUTURE ENHANCEMENT

- Real-time detection of illegal crypto transactions.
- Support for other cryptocurrencies like Ethereum and Litecoin.
- Use of advanced deep learning techniques for better accuracy.
- Integration with banks and crypto exchanges for automatic alerts.
- Better handling of large-scale crypto data.
- Addition of privacy-preserving techniques for secure monitoring.

THANK
YOU!

